We toss a coin 3 times. Let X be the number of heads.

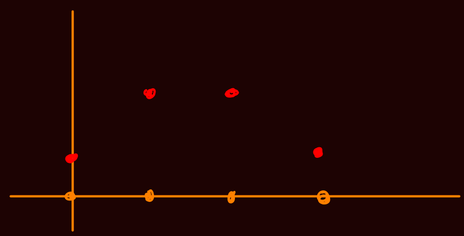
X is called a random variable, $X \in \{0,1,2,3\}$.

$\mathbb{P}(X=0) = \frac{1}{8}$

$\mathbb{P}(X=1) = \frac{3}{8}$ $\qquad \frac{1}{8}+\frac{3}{8}+\frac{3}{8}+\frac{1}{8} = 1$

$\mathbb{P}(X=2) = \frac{3}{8}$ $\qquad$ probability mass function (pmf)

$\mathbb{P}(X=3) = \frac{1}{8}$ $\qquad p_X(x) = p(x)$



We roll a die 2 times. Let X be the sum of the 2 rolls. $X \in \{2,3,\dots,12\}$

x	2	3	4	5	6	7	8	9	10	11	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

What is the probability mass function of the median number drawn in the HUN LOTTERY?

Let that random number be $X \in \{3,4,\dots,87,88\}$.

$$\mathbb{P}(X=x) = \frac{\binom{x-1}{2}\binom{90-x}{2}}{\binom{90}{5}}, \quad x = 3,4,\dots,88$$

Suppose we flip a coin 100 times. Let X be the number of heads. $X \in \{0,1,\dots,100\}$

$$\mathbb{P}(X=x) = \frac{\binom{100}{x}}{2^{100}}, \quad x = 0, 1, \dots, 100$$

$$\mathbb{P}(X=50) = \underbrace{\binom{100}{50} \cdot \frac{1}{2^{100}}}_{\approx 0.079} \approx \underbrace{\frac{1}{\sqrt{50\pi}}}_{\approx 0.0797}$$

If we have $2n$ tosses,

$$\mathbb{P}(X=n) = \binom{2n}{n}\frac{1}{2^{2n}} \sim \frac{1}{\sqrt{n\pi}} \qquad \text{(Stirling-formula)}$$

2 red, 5 white balls in a box.

We draw 3 (without putting back)

X be the number of red balls drawn.

x	0	1	2
$p(x)$	$\frac{\binom{5}{3}}{\binom{7}{3}}$	$\frac{\binom{2}{1}\binom{5}{2}}{\binom{7}{3}}$	$\frac{\binom{2}{2}\binom{5}{1}}{\binom{7}{3}}$

$X \sim \mathrm{Bin}(n, p)$, such that $n \cdot p \approx \lambda > 0$.

Suppose that $n \to \infty$, $p \to 0$ such that $n \cdot p \to \lambda > 0$.

$$\mathbb{P}(X=x) = p(x) = \binom{n}{x} p^x (1-p)^{n-x} \xrightarrow[n\to\infty]{} \frac{\lambda^x}{x!} e^{-\lambda}, \quad e \approx 2.71828\dots$$

$\mathbb{P}(A) = p > 0$.

We perform experiments about A till A happens.

Let X be the number of experiments needed.

$\underbrace{\bar{A}\,\bar{A}\cdots\bar{A}\,A}_{x \text{ times}}$ $\qquad \mathbb{P}(X=x) = (1-p)^{x-1}\cdot p, \quad x = 1, 2, \ldots$